OPTIMIZING 311 WORKER ALLOCATION ACROSS NYC BOROUGHS

Data Cleaning • EDA • Optimization • Sensitivity Analysis

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BUSINESS PROBLEM

NYC must allocate limited 311 workers across boroughs while balancing:

High demand regions	Long service times	Existing backlog	Minimum staffing requirements

The goal is to minimize mismatch between staffing and operational workload

EXECUTIVE SUMMARY

- Brooklyn/Queens account for majority of modeled workload
- Median resolution time better captures workload than mean
- Optimization reallocates workers proportionally to workload
- Sensitivity testing confirms model robustness

METHODOLOGY

1. Data Cleaning & Preprocessing
2. Exploratory Analysis
3. Outlier Detection / Metric Engineering
4. Workload Formulation
5. Optimization Model Development
6. Sensitivity Testing
7. Recommendation Generation

PROJECT OVERVIEW

- Analyzes 311 service request patterns
- Identifies workload drivers (demand, backlog, resolution time)
- Builds a proportional fairness optimization model
- Allocates 1000 workers across boroughs
- Tests robustness with sensitivity analysis
- Uses pandas, numpy, matplotlib.pyplot, seaborn, and PuLP Python libraries
- Data is from NYC Open Data: 311 Service Requests from 2020 to Present
  - Filtered to be from Department of Parks and Recreation (DPR)
  - Filtered to be from 01-01-2022 to 12-31-2025

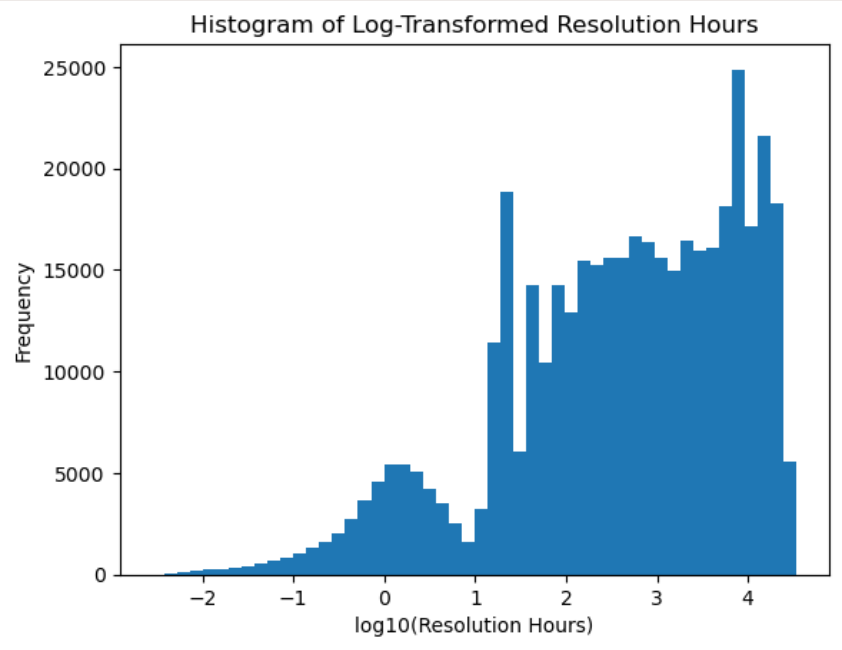
DATASET SUMMARY

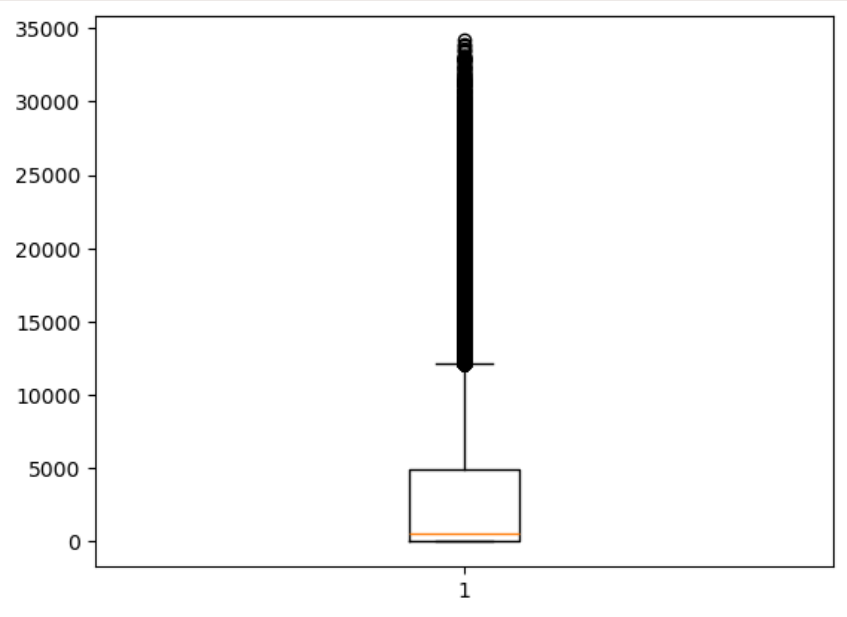
- ~470k cases of 311 New York Requests
- Key fields:
  - Created Date
  - Closed Date
  - Problem (formerly Complaint Type)
  - Borough
- Variables Engineered:
  - Created & Closed Months/Day of Week
  - Resolution Days & Resolution Hours
- Separated into resolvedServiceRequests & unresolvedServiceRequests datasets

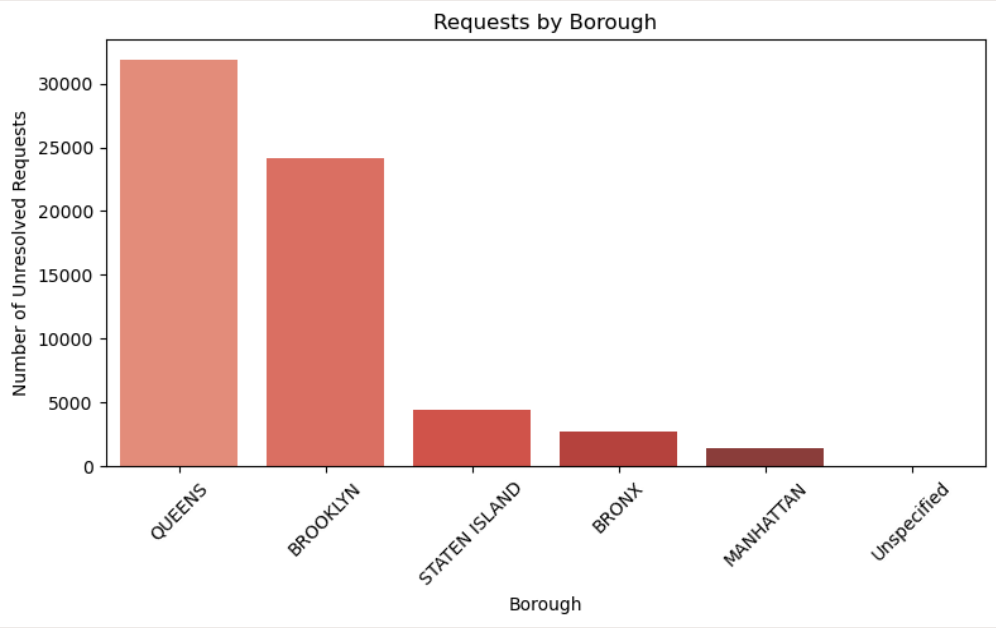
- **Data preparation included:**
  - Removed irrelevant columns
  - Cleaned missing values
  - Missing Borough → *Unspecified*
  - Missing City → Unknown
  - Dropped rows missing Longitude, Latitude, or Incident Zip
  - Parsed date fields
  - Extracted time features (month, day, day of week)

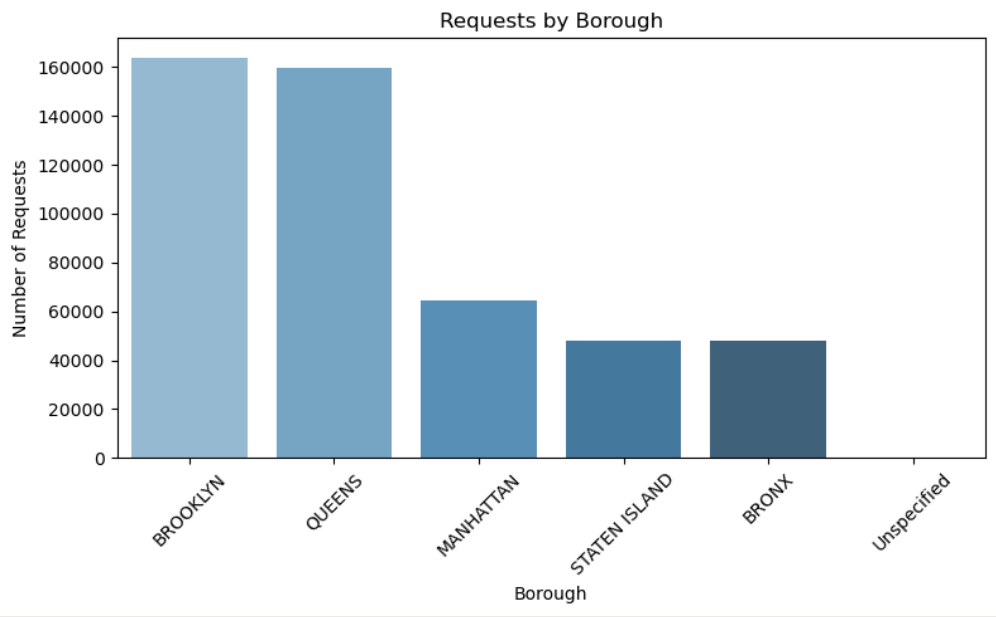
Resolution Hours Outliers

- Resolution Hours had extreme outliers (up to 34,000 hours)
  - Resolution times showed severe right skew, with the 50th percentile at 574 hours and the 99th percentile exceeding 25,000 hours
- The Mean was distorted and the median was more reliable
  - Median– 574
  - Mean - 3754
- Decision: Use median resolution hours per borough





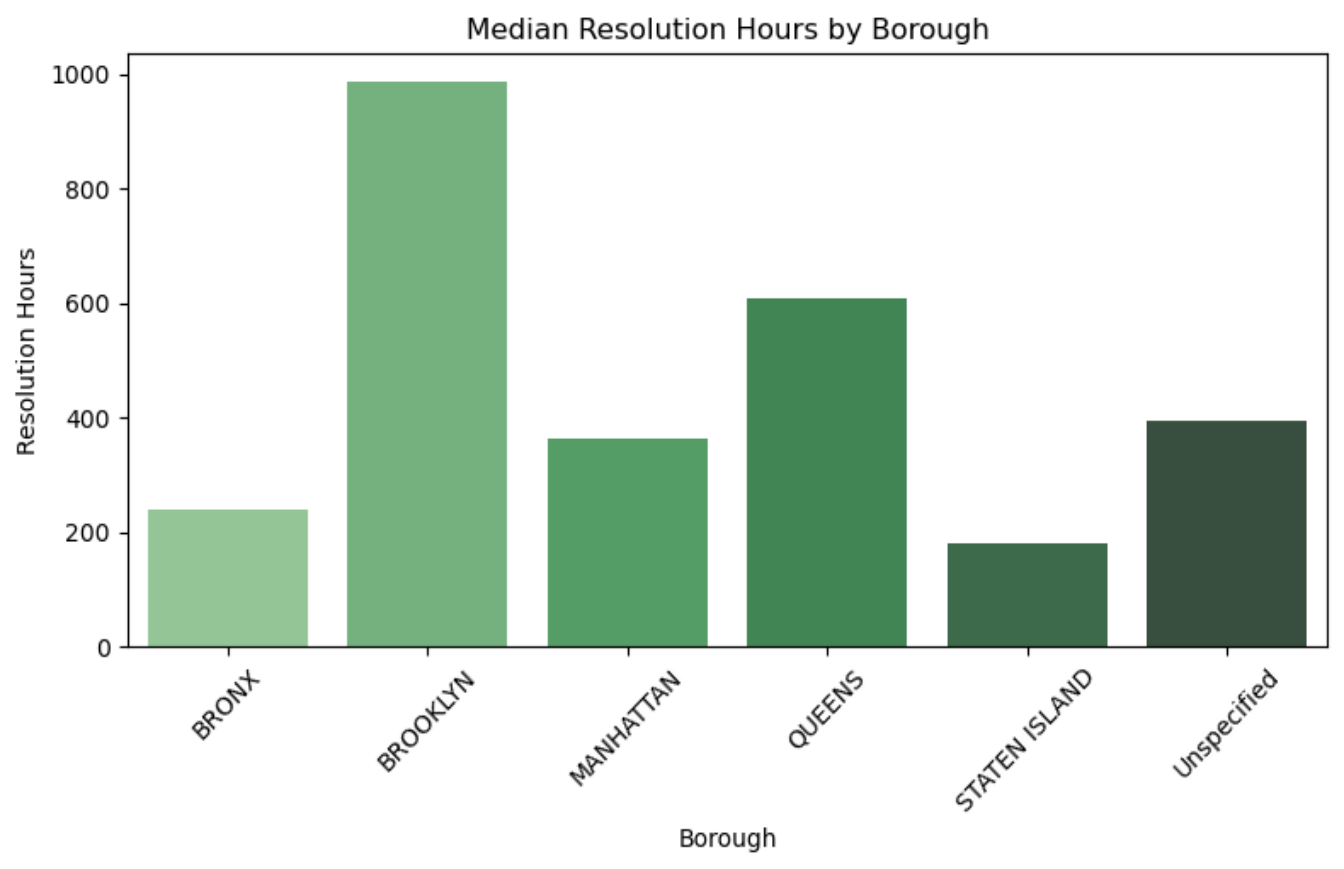




DEMAND & BACKLOG OVERVIEW

- Unspecified has far fewer requests than any borough
- Manhattan has the lowest number of unresolved requests
- Queens and Brooklyn dominate both total requests (demand) and unresolved requests (backlog)
- Brooklyn has the highest total request volume
- Queens has the highest backlog
- Backlog (Red) is included in Demand (Blue)

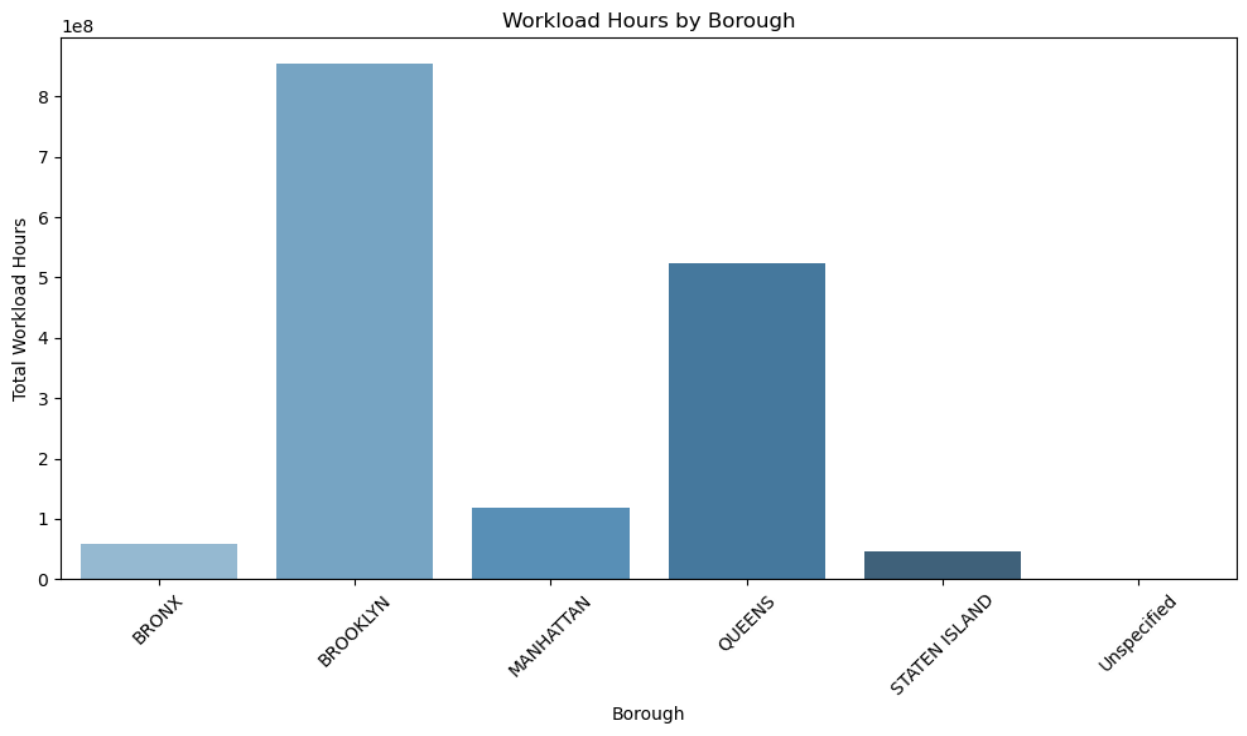
MEDIAN RESOLUTION HOURS BY BOROUGH



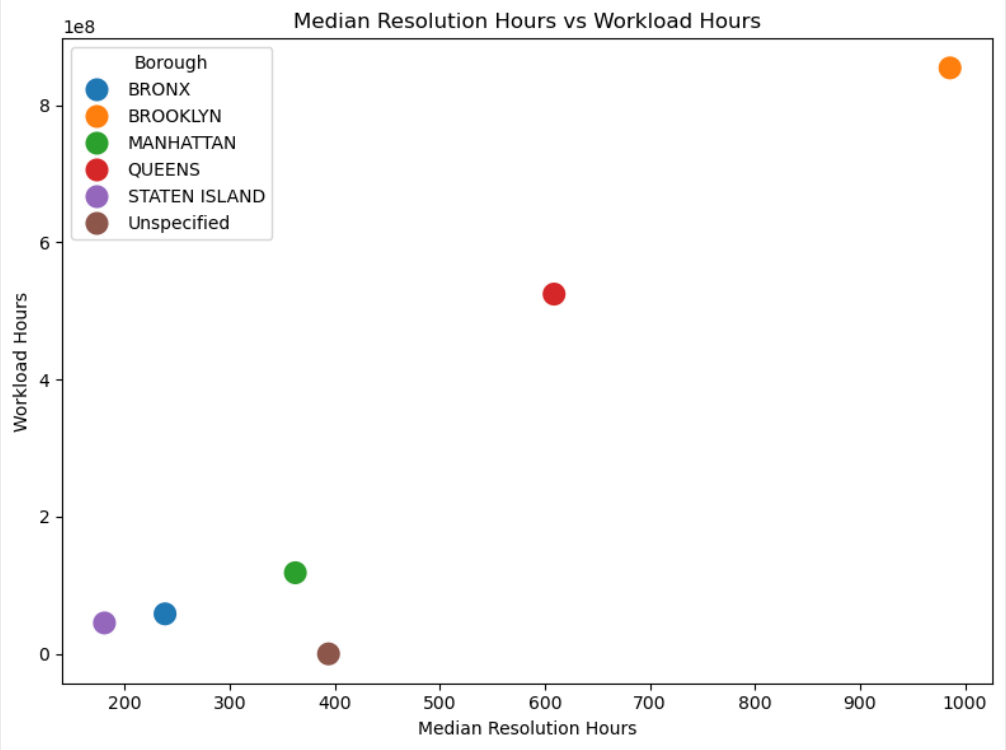
- Brooklyn has the longest median resolution time
- Staten Island has the shortest median resolution time
- Queens is the second slowest
- Brooklyn and Queens both have high demand AND slow resolution, creating the largest workload burden

WORKLOAD

- Workload = demand + 2 · backlog
- Workload Hours = workload · median Resolution Hours



- Brooklyn has the highest Median Resolution Time and the highest Workload Hours
- Workload depends on volume and difficulty
  - While unspecified is difficult it has a low volume of cases
  - Brooklyn is both high volume and difficult
  - Queens has less difficulty than Brooklyn while being similar in volume resulting in less Workload Hours



WORKER ALLOCATION MODEL COMPARISON

Model	Description	Strengths	Weaknesses	Result
Mean Resolution Model	Uses mean service time in workload calculation	Simple baseline	Highly distorted by outliers	Rejected
Median Resolution Model	Replaces mean with median	Robust to skew/outliers	Better reflects typical workload	Selected
Forecast-Based Model	Incorporates projected demand trends	Forward-looking	Similar results due to stable trend	Supplemental
Heuristic Score Model	Weighted scoring approach	Interpretable/simple	Less mathematically rigorous	Benchmark Only

WHY MEDIAN-BASED MODEL

- Mean resolution time heavily skewed by extreme outliers
- Forecast adjustments produced minimal allocation changes
- Heuristic model lacked optimization rigor
- Median-based workload model balanced robustness, interpretability, and fairness

OPTIMIZATION MODEL

Each borough should receive workers proportional to its workload

Slack variables allow controlled deviation from ideal proportional allocation

Minimum staffing constraint ensures operational feasibility (10 Workers per borough)

Distributing 1000 workers

The goal is to minimize the proportional mismatch across the boroughs

Achieved by minimizing the sum of over and under deviations

MATHEMATICS FORMULA

Objective Formula

$$\min \sum_{b} (Over_b + Under_b)$$

Subject to

$$\frac{Workload_b}{TotalWorkload} - \frac{Workers_b}{TotalWorkers} = Over_b - Under_b$$

$$Workers_b \geq 10$$

$$\sum_{b} Workers_b = 1000$$

Decision Variables

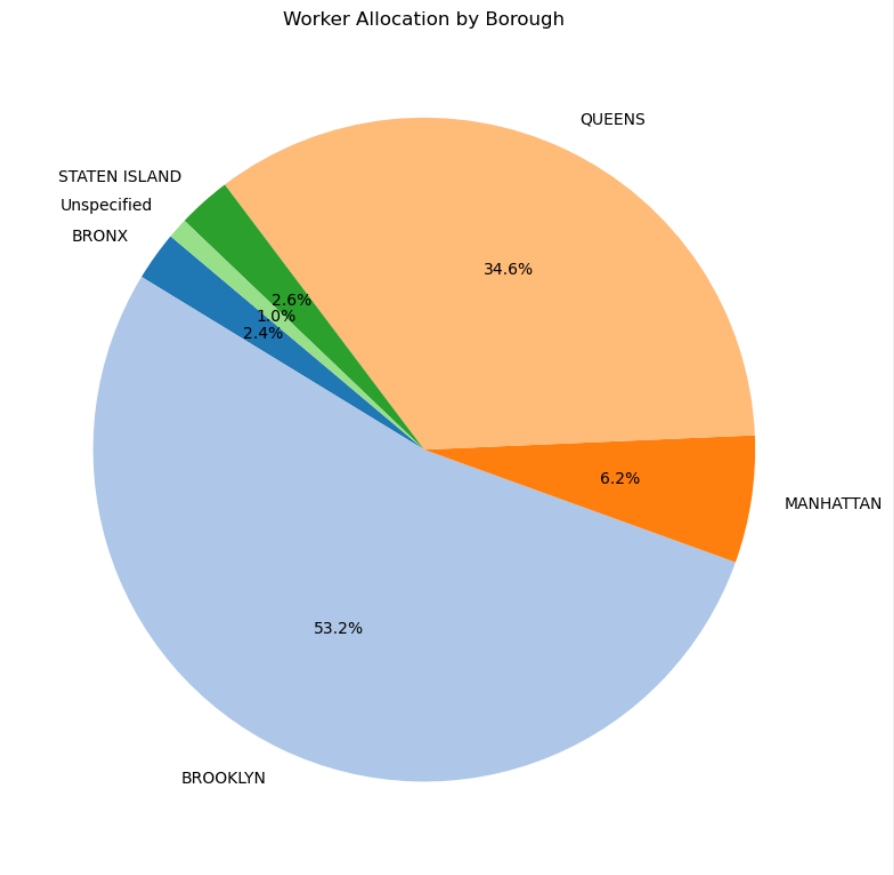
- Workers_b : Workers assigned to borough b

Objective

- Minimize staffing-workload mismatch

Constraints

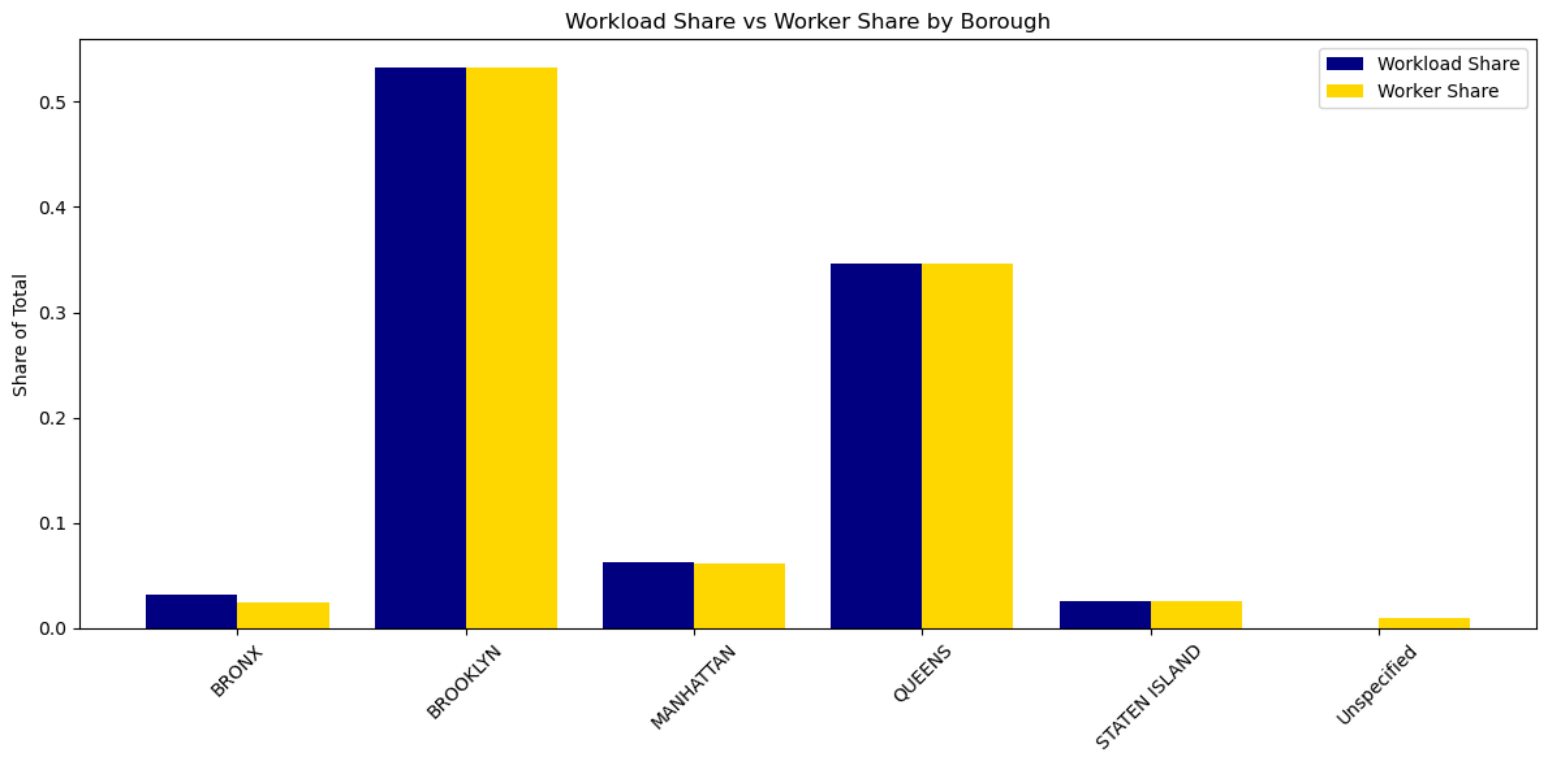
- Maintain proportional fairness
- Enforce minimum staffing
- Preserve total workforce count



WORKER ALLOCATION

Bronx received 24 Workers
Brooklyn received 532 Workers
Manhattan received 62 Workers
Queens received 346 Workers
Staten Island received 26 Workers
Unspecified received the minimum of 10 Workers

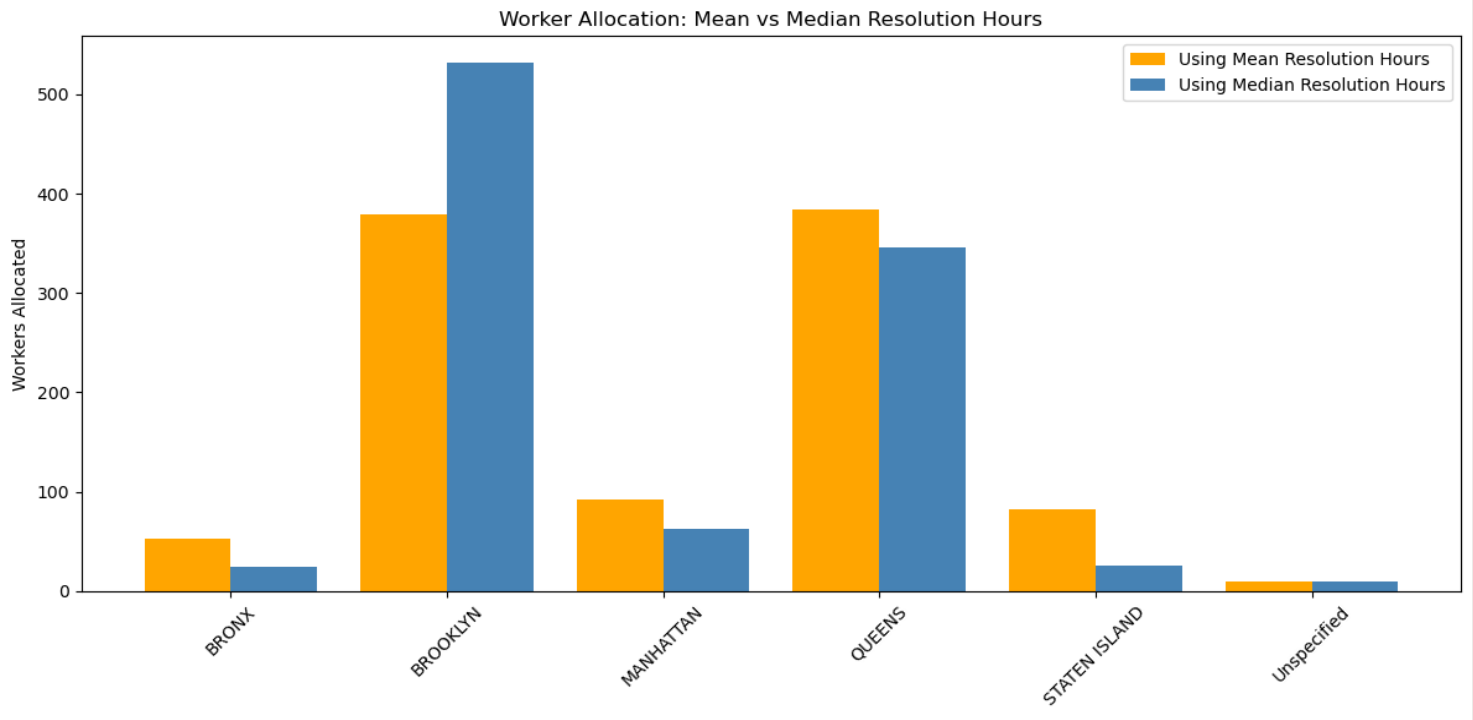
- Brooklyn and Queens receive the largest share of workers because they have the highest workload hours
- Unspecified receives only 10 workers, which is the minimum staffing requirement
- The allocation reflects both demand and resolution time, not just number of cases



FAIRNESS CHECK

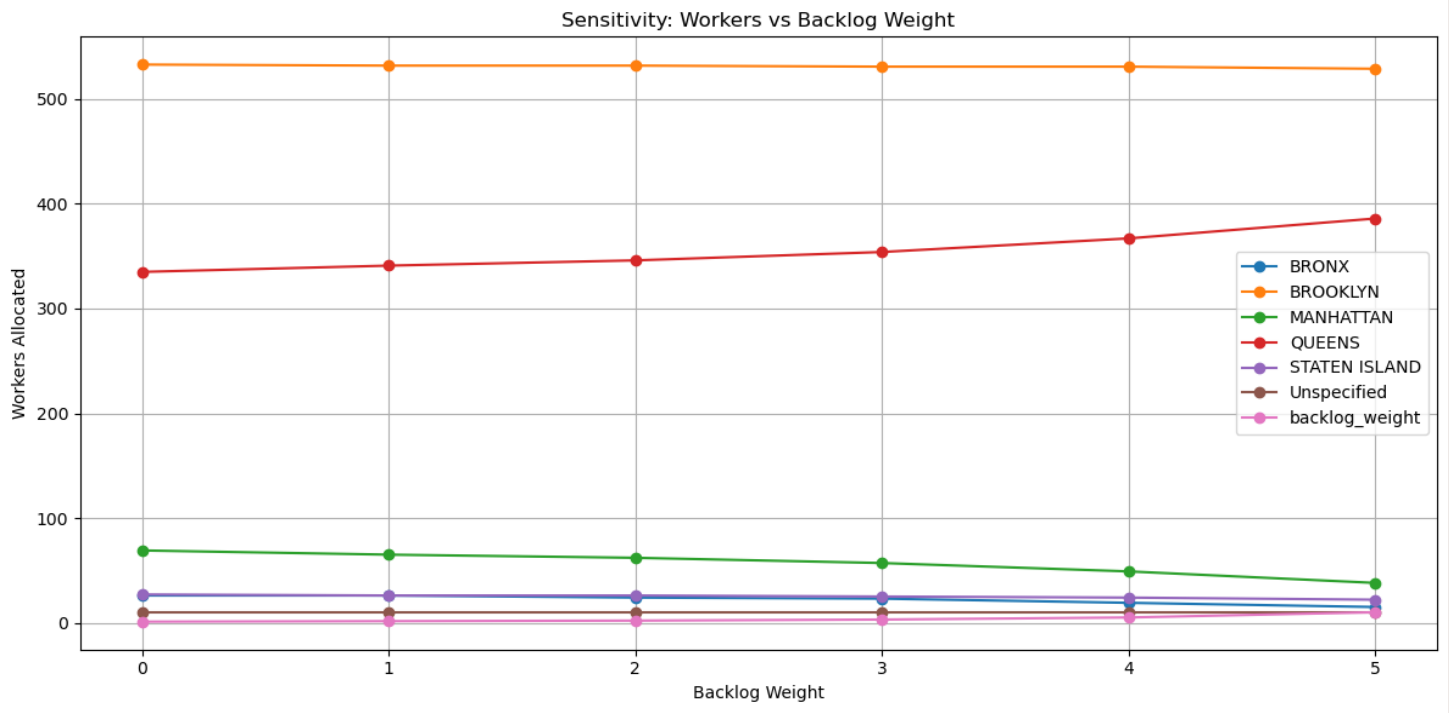
The workload and worker shares are nearly identical except for unspecified requiring some workers

IMPACT OF ROBUST RESOLUTION-TIME MODELING



- Using the Median Resolution time the allocation is more distributed towards where most of the work is coming from (Brooklyn and Queens)
- The Median Resolution time allows workload hours to influence the result
- Brooklyn is the only borough to lose workers from using Median to Mean Resolution time
- Unspecified doesn't change

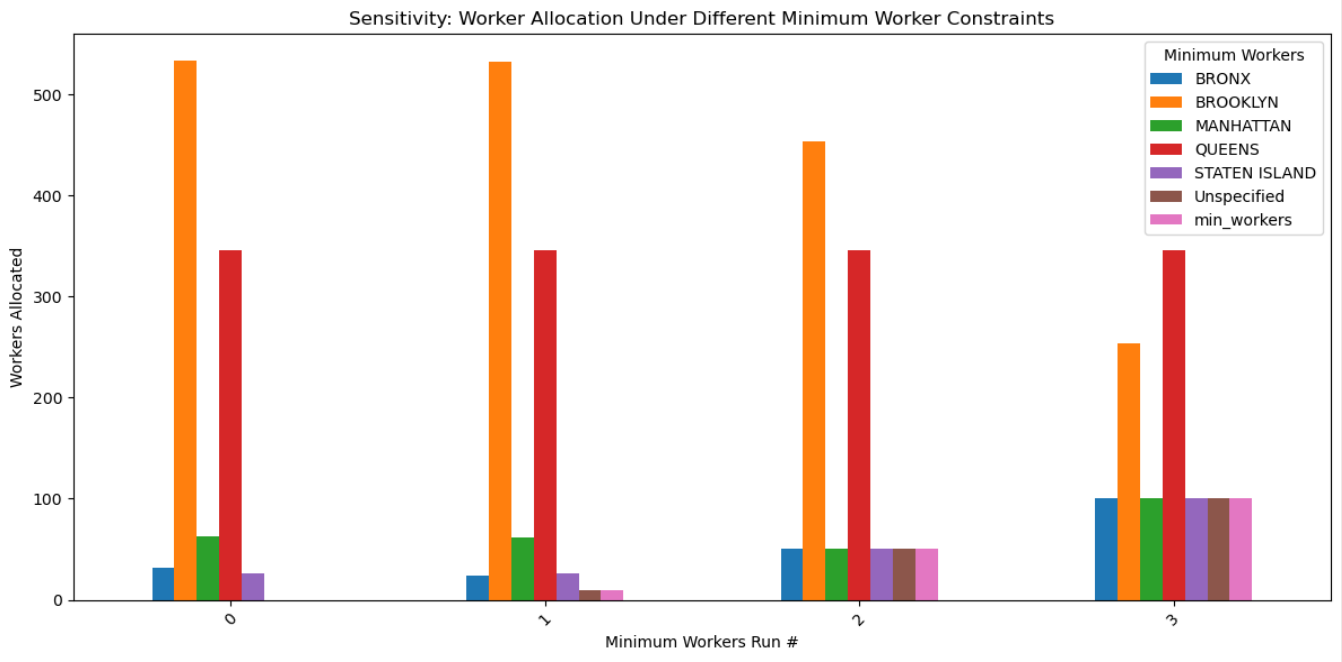
SENSITIVITY ANALYSIS - BACKLOG WEIGHT



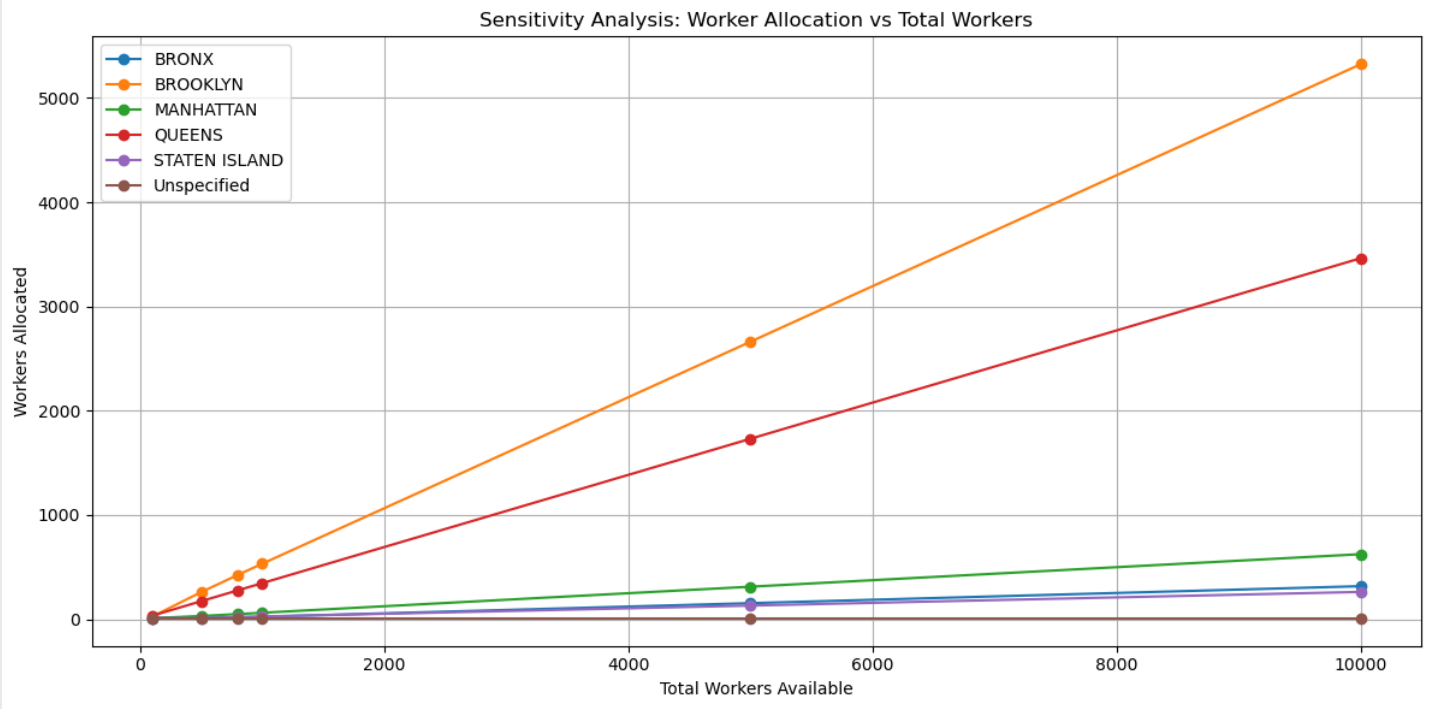
- Unspecified doesn't change
- Queens has the biggest change between each change
- Queens increases as the weight on Backlog increases
  - Positive relationship
- Manhattan, Brooklyn, and Staten Island decrease as the weight on Backlog increases
  - Negative relationship

SENSITIVITY ANALYSIS – MINIMUM WORKER

- Minimum values of 0, 10, 25, 50, and 100 were used
- The number of workers for Queens does not change
- Unspecified is always allocated the minimum number of workers

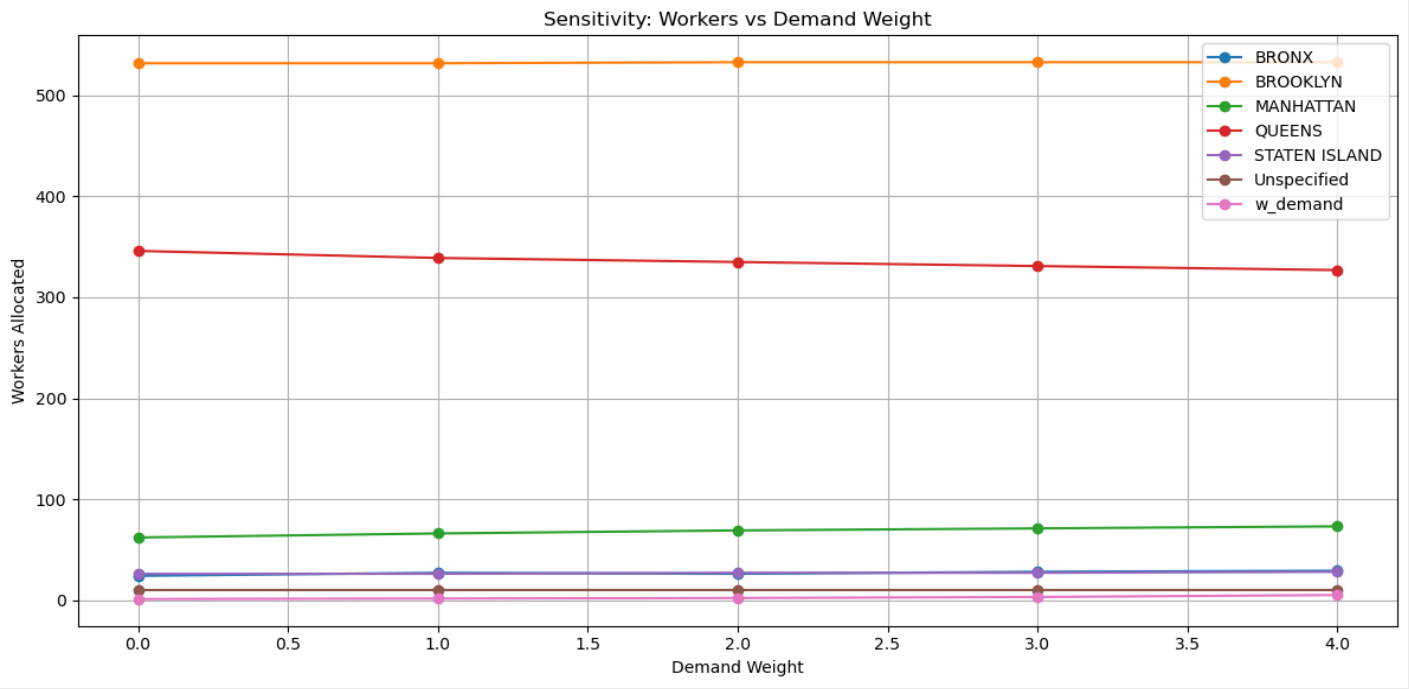


SENSITIVITY ANALYSIS – WORKFORCE CAPACITY



- Brooklyn and Queens scale rapidly as total workers increase
- Manhattan, Bronx, and Staten Island grow more gradually
- Unspecified stays fixed at 10 workers due to the minimum staffing rule
- Allocation stays proportional to workload across all scenarios
- The model behaves consistently even when total workforce changes dramatically
- Bronx starts to grow faster than Staten Island and gets more workers after 1000 total workers

SENSITIVITY ANALYSIS – DEMAND WEIGHT



- Brooklyn and Staten Island slowly increases as Demand weight increases
  - 1 worker
- Queens decreases as Demand weight increases
- Manhattan and Bronx follow similar increasing lines as Demand Weight increases
- Unspecified doesn't change

KEY FINDINGS

- Worker demand is highly concentrated in Brooklyn and Queens, which account for the majority of modeled workload
- Median resolution time provides a significantly more stable workload estimate than mean due to extreme outliers
- Backlog weighting materially affects borough allocation, particularly for Queens
- Minimum staffing constraints primarily affect low-volume regions like Unspecified and Staten Island
- Model stays stable across workforce capacity scenarios, suggesting robustness

MODEL ASSUMPTIONS

- All workers are assumed to have equal productivity
- Workers are assumed interchangeable across boroughs
- Travel time between service locations is not modeled
- Resolution time is assumed representative of workload difficulty
- Historical demand is assumed indicative of future demand
- Complaint type specialization is not incorporated

RECOMMENDATIONS

01

Prioritize staffing increases in Brooklyn and Queens under current demand conditions

02

Continue using median-based service time metrics for operational planning

03

Reevaluate backlog weighting periodically based on service-level objectives

04

Implement rolling demand forecasts to dynamically update worker allocations

05

Segment by complaint type in future models for more granular staffing

CONCLUSIONS

- Developed a mathematically grounded workforce allocation framework for NYC 311 operations
- Combined demand, backlog, and service-time metrics into a workload model
- Applied linear programming optimization to allocate staffing proportionally and fairly
- Validated model robustness through multiple sensitivity analyses
- Produced an extensible framework for future operational planning enhancements

FUTURE WORK

Forecast	Forecast borough demand using time-series models
Add	Add complaint-type-specific worker specialization constraints
Incorporate	Incorporate geographic routing/travel time optimization
Model	Model dynamic reallocation over multiple time periods
Extend	Extend to stochastic/uncertain demand scenarios
Optimize	Optimize allocation by complaint type rather than borough